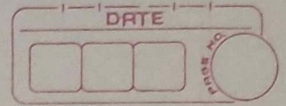


SQL DB



What is Database -

It is a collection of data in a format that can be easily Accessed.

SQL V/s NoSQL -

SQL -

Relational DB (Data stored in tables)

eg - MySQL, Oracle, PostgreSQL etc

NoSQL -

Non Relational Database (Data stored in Document / Key-Val / Graphs etc)

eg - MongoDB, Cassandra, Neo4j etc



SQL (Structured Query language)

SQL is a programming lang. used to interact with Relational DB.

our 1st DB

CREATE DATABASE College;

DROP DATABASE College; — For Deleting DB.

USE College; — For creating Tables / working on DB.

Syntax -

```
CREATE TABLE table_name (  
    Column - name 1 datatype Constraint,  
    Column - name 2 datatype Constraint,  
    Column - name 3 datatype Constraint,  
);
```

Eg -

```
CREATE DATABASE College;  
USE College;
```

```
CREATE TABLE Student (  
    rollno INT,  
    name VARCHAR(30),  
    age INT  
);
```

```
INSERT INTO Student  
VALUES
```

```
(101, "adam", 12),  
(102, "bob", 14);
```

```
SELECT * FROM Student;
```


Database Queries

CREATE DATABASE db-name;

CREATE DATABASE IF NOT EXISTS db-name;

DROP DATABASE db-name;

DROP DATABASE IF EXISTS db-name;

SHOW DATABASES;

SHOW TABLES;

Table Queries

Create - For creating Table

insert - For inserting data into Table

update - For updating data

Alter - For Altering schema, changing Columns

truncate - Delete all data & keep empty

Delete - Deleting Table

Constraints - Rules For data in the Table

NOT NULL - Columns cannot have a null value

UNIQUE - all values in column are different

DEFAULT - sets the default value of a column

CHECK - it can limit the values allowed in column

PRIMARY KEY-

makes a column unique & not null but used only for one.

```
CREATE TABLE temp (
    id int not null,
    PRIMARY KEY (id)
);
```

FOREIGN KEY-

Prevent Actions that would destroy links between tables

```
CREATE TABLE temp (
    cust_id int,
    FOREIGN KEY (cust_id) references customer (id)
);
```

What are keys?

Keys are special columns in the Table

Primary Key-

It is a column (or set of columns) in a Table that uniquely identifies each row (or unique id). There is only 1 PK & it should be not null.

Foreign Key-

A Foreign Key is a column (or set of columns) in a Table that refers to the primary key.

in another Table FKs can have duplicate & Null values.

There can be multiple FKs.

Table Queries

Insert Into Table

Syntax -

INSERT INTO table-name

(Colname 1, Colname 2);

VALUES

(Col1-v1, Col2-v1),

(Col1-v2, Col2-v2);

Eg -

INSERT INTO user

(id, age, name, email, Followers, Following)

VALUES

(1, 14, "adam", "adm@yahoo.in", 123, 145),

(2, 15, "bob", "bob@bob.in", 157, 200);

Select Command

Select & Show Data From the DB

Syntax

SELECT Col 1, Col 2 **FROM** table-name;

Syntax (to show all)

SELECT * FROM table-name;

Where clause

To define conditions

SELECT col1, col2 **FROM** table-name
WHERE conditions;

Eg -

SELECT name, followers
FROM user
WHERE followers $\geq$ 200;

Another operators -

Arithmetic operators : +, -, *, /, %

Comparison operators : =, !=, >, <, >=, <=

Logical operators : AND, OR, NOT, IN, BETWEEN

Bitwise operators : &, |

Frequently used operators.

AND (to check for both conditions to be true)

OR (to check for one of the conditions to be true)

BETWEEN (selects for a given Range)

IN (matches any value in the list)

NOT (to negate the given condition)

Order by clause -

To Sort in Ascending (ASC) or descending order (DESC)

```
SELECT col1, col2 FROM table-name  
ORDER BY col1-name (S) ASC;
```

Limit clause -

Sets an upper limit on number of (tuples) rows to be Returned.

```
SELECT col1, col2 FROM table-name  
LIMIT number -
```

Aggregate Functions -

Aggregate Functions Perform a Calculation on a set of values, and return a single value.

- COUNT()
- MAX()
- MIN()
- SUM()
- AVG()

Eg -

```
SELECT max(marks)  
FROM Students;
```


Group by clause -

Groups rows that have the same values into summary rows.

It collects data from multiple records & groups the result by one or more column.

```
SELECT col1, col2
```

```
FROM table-name
```

```
GROUP BY col-name(s);
```

* Generally we use group by with some Aggregation function.

Having clause -

Similar To where i.e. applies some conditions on rows But it is used when we want to apply any condition after grouping.

```
SELECT col1, col2
```

```
FROM table-name
```

```
GROUP BY col-name(s)
```

```
HAVING condition;
```

- where is for the table, HAVING is for a group
- Grouping is necessary for HAVING.

General order - (command for creating table)

SELECT Column(s)

FROM table-name

WHERE Condition

GROUP BY Column(s)

HAVING Condition

ORDER BY Col (s) ASC;

Table Queries

UPDATE (To update Existing Row)

UPDATE table-name

SET Col1 = Val1, Col2 = Val2

WHERE Condition;

DELETE (To Delete existing row)

DELETE FROM table-name

WHERE Condition;

Ex -

DELETE FROM user

WHERE age = 14;

ALTER (To change the Schema)

Add Column

ALTER TABLE table-name

ADD COLUMN column-name datatype constraint;

Drop Column

ALTER TABLE table-name

DROP COLUMN column-name;

RENAME TABLE

ALTER TABLE table-name

RENAME TO new-table-name;

TRUNCATE (to Delete table's data)

TRUNCATE TABLE table-name;